Quiz 1, 2021/2022 ME 265 THERMODYNAMICS I

INSTRUCTIONS

Select the correct ANSWER and check your answer appropriately. Questions 1 -40 will attract one mark for each question answered correctly. Questions 41 – 60 will attract 3 marks for each question answered correctly.

TEMPERATURE BASICS (1 mark for each correct answer for questions 1 - 40)

1. Which of the following were used as fixed points before 1954?
a) The ice point
b) The steam point
c) All of the mentioned
d) None of the mentioned

2. All gases and vapours approach ideal gas behaviour at?
a) High pressure and high density
b) Low pressure and low density
c) High pressure and low density
d) Low pressure and high density

3. Celsius temperature of the triple point of water is (in degree Celsius)?
a) -0.00
b) 0.00
c) 0.01
d) None of the mentioned

4. A real gas behaves as an ideal gas when?
a) Temperature approaches zero
b) Pressure approaches zero
c) Both temperature and pressure approach zero
d) None of the mentioned

5. Optical method is adopted for measuring temperatures higher than the gold point?
a) True
b) False

FIRST LAW

6. Energy has different forms which include
a) heat
b) work
c) all of the mentioned
d) none of the mentioned

7. Which of the following represents the energy in storage?
a) heat
b) work
c) internal energy
d) none of the mentioned

8. The expression (ΣW) cycle = (ΣQ) cycle applies only to systems undergoing cycles.
a) True
b) false

9. Which of the following can be considered as the definition of energy?
a) Q=ΔE+W
b) Q-W=ΔE
c) first law of thermodynamics
d) all of the mentioned

ENERGY

10. The specific energy, e=E/m is an extensive property.
a) True
b) false

11. (m*g*z) gives the
a) macroscopic kinetic energy
b) microscopic kinetic energy
c) macroscopic potential energy
d) microscopic potential energy

12. In an ideal gas there are no intermolecular forces of attraction and repulsion, and the internal energy is a function of temperature only.
a) True
b) false

13. A perpetual motion machine of first kind
a) is a fictitious machine
b) can supply mechanical work without dissipating energy
c) violates first law
d) all of the mentioned

14. The limitation of the first law is
a) does not indicate the possibility of a spontaneous process proceeding in a definite direction
b) it assigns a quality to different forms of energy
c) indicates the direction of any spontaneous process
d) none of the mentioned

ENTHALPY

15. In a constant volume process, internal energy change is equal to
a) heat transferred
b) work done
c) zero
d) none of the mentioned

16. Enthalpy is an intensive property of a system.
a) True
b) false

17. The enthalpy of an ideal gas depends only on the temperature.
a) True
b) false

18. Change in enthalpy of a system is due to heat supplied at
a) constant volume
b) constant pressure
c) both at constant volume and pressure
d) none of the mentioned

CONTROL VOLUME SYSTEMS

19. The specific heat of a substance at constant volume is defined as the rate of change of ___ with respect to ___

a) specific internal energy, temperature
b) work, pressure
c) specific internal energy, pressure
d) heat, temperature

20. Specific heat of a substance at constant volume is a property of the system.

a) True
b) false

21. The heat capacity at constant pressure Cp is:

a) m/cp
b) cp/m
c) m×cp
d) none of the mentioned

22. When there is mass transfer across the system boundary, the system is called

a) isolated system
b) closed system
c) open system
d) none of the mentioned

23. Steady flow means that the rates of flow of mass and energy across the control surface

a) varies
b) remains constant
c) depends on the control surface
d) none of the mentioned

FIRST LAW APPLIED TO FLOW PROCESS – STEADY FLOW PROCESS

24. In a flow process, the work transfer may be of which type?

a) External work
b) flow work
c) all of the mentioned
d) none of the mentioned

25. Which of the following represents the steady flow energy equation?
a) $Q \pm Wx=(h_2-h_1)-(V_2-V_1)(V_2-V_1)/2+g(Z_2-Z_1)$
b) $Q \pm Wx=(h_2-h_1)+(V_2-V_1)(V_2+V_1)/2+g(Z_2+Z_1)$
c) $Q \pm Wx=(h_2-h_1)-(V_2-V_1)(V_2+V_1)/2+g(Z_2-Z_1)$
d) $Q \pm Wx=(h_2-h_1)+(V_2-V_1)(V_2+V_1)/2+g(Z_2-Z_1)$

26. In the differential form, the SFEE becomes
a) $\delta Q \pm \delta W = dh + VdV - gdZ$
b) $\delta Q \pm \delta W = dh + VdV + gdZ$
c) $\delta Q \pm \delta W = dh - VdV - gdZ$
d) $\delta Q \pm \delta W = dh - VdV + gdZ$

27. When more than one fluid stream enters or leaves the control volume, which type of balance is taken?
a) Mass balance
b) energy balance
c) mass balance and energy balance
d) none of the mentioned

28. The flow work is the displacement work done by the fluid and is given by
a) -pvdm
b) pvdm
c) pvdm or -pvdm depending on whether it is inlet or exit
d) none of the mentioned

FIRST LAW APPLIED TO FLOW PROCESS – STEADY FLOW PROCESSES

29. What does a nozzle do?
a) decreases the velocity of a fluid at the cost of its pressure gain
b) increases the velocity of a fluid at the cost of its pressure drop
c) increases the velocity of a fluid and also its pressure
d) none of the mentioned.

30. Fluid flow through which of the following throttles the flow?
a) Partially opened valve
b) orifice
c) porous plug
d) all of the mentioned

31. In a throttling device, what do we get as SFEE when changes in PE and KE are taken zero?
a) dQ/dm≠0
b) dW/dm≠0
c) h1=h2
d) none of the mentioned

32. For a turbine, it is seen that work is done by the fluid at the expense of its enthalpy.
a) True
b) false

33. A heat exchanger is a device in which heat is transferred from one fluid to another.
a) True
b) false

34. The Bernoulli equation is restricted to _____ fluids but the SFEE is valid for _____ fluids as well.
a) Viscous compressible, frictionless incompressible
b) frictionless incompressible, viscous compressible
c) viscous incompressible, frictionless compressible
d) none of the mentioned

35. When a body A is in thermal equilibrium with a body B, and also separately with a body C, then B and C will be in thermal equilibrium with each other.
a) True
b) False

36. What is the standard fixed point of thermometry?
a) The ice point
b) The steam point
c) The triple point of water
d) None of the mentioned

37. The value of ratio of the steam point temperature to the ice point temperature is?
a) 1.466
b) 1.266
c) 1.166
d) 1.366

38. Which of the following is chosen as the standard thermometric substance?
a) Gas
b) Liquid
c) Solid
d) All of the mentioned

39. The temperature interval from the oxygen point to the gold point is divided into how many parts?
a) 2
b) 3
c) 4
d) 1

40. Work input is directly proportional to heat and the constant of proportionality is called
a) joule's equivalent
b) mechanical equivalent of heat
c) all of the mentioned
d) none of the mentioned

BASIC CONCEPTS (3 marks for each correct answer for questions 41- 60)

41. A 100 kg piston/cylinder with a cross-sectional area of 0.01 m^2 is resting on the stops. With an outside pressure of 100 kPa, what should be the water pressure to lift the piston?
a) 178.1 kPa
b) 188.1 kPa
c) 198.1 kPa
d) 208.1 kPa

42. A 5 m long vertical tube having cross sectional area 200 cm^2 is placed in a water. It is filled with 15 °C water, with the bottom closed and the top open to 100 kPa atmosphere. How much water is present in tube?
a) 99.9 kg
b) 109.9 kg
c) 89.9 kg
d) 79.9 kg

43. Find the pressure of water at 200 °C and having specific volume of 1.5 m^3 /kg.
a) 141.6 kPa
b) 111.6 kPa
c) 121.6 kPa
d) 161.6 kPa

44. A piston-cylinder device initially contains air at 150 kPa and 27 °C. At this state, the volume is 400 litres. The mass of the piston is such that a 350 kPa pressure is required to move it. The air is now heated until its volume has doubled. Determine the final temperature.
a) 1400 K
b) 400 K
c) 500 K
d) 1500 K

45. A 100 m tall building receives superheated steam at 200 kPa at ground and leaves as saturated vapour from the top at 125 kPa by losing 110 kJ/kg of heat. What should be the minimum inlet temperature at the ground of the building so that no steam will condense inside the pipe at steady state?
a) 363.54 °C
b) 263.54 °C
c) 163.54 °C
d) none of the mentioned

46. Find the change in specific internal energy for carbon dioxide between 600 K and 1200 K for a constant Cv = 0.657 kJ/kg K.
a) 291.2 kJ/kg
b) 394.2 kJ/kg
c) 491.2 kJ/kg
d) 591.2 kJ/kg

47. A sealed rigid vessel has volume of 1 m^3 and contains 2 kg of water at 100 °C. The vessel is now heated. If a safety pressure valve is installed, at what pressure should the valve be set to have a maximum temperature of 200 °C?
a) 431.31 kPa
b) 531.31 kPa
c) 631.31 kPa
d) 731.31 kPa

POLYTROPIC PROCESS – 1

48. The piston/cylinder contains carbon dioxide at 300 kPa, with volume of 0.2 m^3 and at 100 °C. Mass is added at such that the gas compresses with $PV^{1/2}$ = constant to a final temperature of 200 °C. Determine the work done during the process.

a) -80.4 kJ
b) -40.4 kJ
c) -60.4 kJ
d) -50.4 kJ

49. A mass of 1kg of air contained in a cylinder at 1000 K, 1.5 MPa, expands in a reversible adiabatic process to 100 kPa. Calculate the work done during the process using Constant specific heat.

a) 286.2 kJ
b) 386.2 kJ
c) 486.2 kJ
d) 586.2 kJ

50. A cylinder/piston contains 1kg methane gas at 100 kPa, 20°C. The gas is compressed reversibly to a pressure of 800 kPa. Calculate the work required if the process is polytropic, with exponent n = 1.15 and $R_{methane}$ = 0.51835 kJ/kg K

a) -315.7 kJ
b) -415.7 kJ
c) -515.7 kJ
d) -615.7 kJ

51. Consider air in a cylinder volume of 0.2 L at 7 MPa, 1800K. It now expands in a reversible polytropic process with exponent, n = 1.5, through a volume ratio of 8:1. Calculate the work for the process.

a) 1.61 kJ
b) 1.71 kJ
c) 1.81 kJ
d) 1.91 kJ

52. A cylinder/piston contains 100 L of air at 25 °C, 110 kPa. The air is compressed in a reversible polytropic process to a final state of 200 °C, 800 kPa. Assume the heat transfer is with the ambient at 25 °C. Find the work done by the air.

a) -11.28 kJ
b) -21.28 kJ
c) -31.28 kJ
d) -41.28 kJ

53. A piston/cylinder contains air at 100 kPa, 300 K. A reversible polytropic process with n = 1.3 brings the air to 500 K. Any heat transfer if it comes in is from a 325 °C reservoir and if it goes out it is to the ambient at 300 K. Find the specific work.
a) -171.3 kJ/kg
b) -181.3 kJ/kg
c) -191.3 kJ/kg
d) -201.3 kJ/kg

POLYTROPIC PROCESS – 2

54. A cylinder/piston contains air at 100 kPa and 20°C with a V=0.3 m^3. The air is compressed to 800 kPa in a reversible polytropic process with n = 1.2, after which it is expanded back to 100 kPa in a reversible adiabatic process. Find the net work.
a) -174.6 kJ/kg
b) -154.6 kJ/kg
c) -124.6 kJ/kg
d) -194.6 kJ/kg

55. A gas initially at 500 °C, 1 MPa is contained in a piston-cylinder arrangement with an initial volume of 0.1 m^3. It is then slowly expanded according to the relation PV = constant until a final pressure of 100 kPa is attained. Determine the work for this process.
a) 200.3 kJ
b) 210.3 kJ
c) 220.3 kJ
d) 230.3 kJ

56. Air goes through a polytropic process from 325 K, 125 kPa to 500 K, 300 kPa. Find the specific work in the process.
a) -51.8 kJ/kg
b) -61.8 kJ/kg
c) -71.8 kJ/kg
d) -81.8 kJ/kg

57. A piston-cylinder contains 0.1 kg air at 400 K, 100 kPa which goes through a polytropic compression process (n = 1.3) to a pressure of 300 kPa. How much work has been done by air in the process?
a) -9.04 kJ
b) -10.04 kJ
c) -11.04 kJ
d) -12.04 kJ -477kJ correct ans

58. A balloon behaves according to the equation $P = (C2)\ xV^{1/3}$, C2 = 100 kPa/m. The balloon is blown up with air from a volume of 1 m^3 to a volume of 3 m^3. Find the work done by the air assuming it is at 25°C.
a) 219.5 kJ
b) 229.5 kJ
c) 239.5 kJ
d) 249.5 kJ

59. Consider a piston-cylinder with 0.5 kg of R-134a as saturated vapour at -10 °C. It is compressed to a pressure of 500 kPa in a polytropic process with n = 1.5. Determine the work done during the process.
a) -6.07 kJ
b) -7.07 kJ
c) -8.07 kJ
d) -9.07 kJ

60. A piston-cylinder contains water at 3 MPa, 500 °C. It is cooled in a polytropic process to 1 MPa, 200°C. Find the specific work in the process.
a) 155.2 kJ/kg
b) 165.2 kJ/kg
c) 175.2 kJ/kg
d) 185.2 kJ/kg

Bonus Quiz

F. K. FORSON